

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$$H_0: p_1 = p_2 = p_3 = 1/3$$

H_a : At least one entering 的概率与其他不同

$$n = 83 + 61 + 56 = 200$$

$$E = 200 \times 1/3 = 66.67$$

$$\chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67} \approx 6.19$$

$$df = k - 1 = 2$$

$$\chi^2_{0.05, 2} = 5.99$$

$$\therefore 6.19 > 5.99 \rightarrow \text{Reject } H_0$$

$$\hat{p} = \frac{83}{200} = 0.415$$

$$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.415 \times 0.585}{200}} \approx 0.0348$$

$$C.I. = \hat{p} \pm z \times SE \\ = 0.415 \pm 1.96 \times 0.0348$$

$$\rightarrow (0.3468, 0.4832)$$

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.¹ In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$$H_0: p_1 = p_2 = \dots = p_7 = 1/7$$

H_a : 一周内发生 heart 的概率不完全相同

$$n = 200$$

$$E = \frac{200}{7} \approx 28.57$$

$$\chi^2 = \frac{(24-28.57)^2}{28.57} + \frac{(36-28.57)^2}{28.57} + \frac{(27-28.57)^2}{28.57} + \dots + \frac{(29-28.57)^2}{28.57} \approx 3.629$$

$$\alpha = 0.05, \quad k - 1 = 6$$

$$\chi^2_{0.05, 6} = 12.592$$

$$\therefore 3.629 < 12.592 \rightarrow \text{Reject } H_0$$

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large
Fatal	67	26	16
Not Fatal	128	63	46
	195	89	62

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0 : 事故率与汽车尺寸 independent

H_a : 有关

$$E_{1,1} = \frac{109 \times 195}{346} = 61.43$$

$$E_{2,1} = \frac{27 \times 195}{346} = 15.37$$

$$E_{1,2} = \frac{109 \times 89}{346} = 28.04$$

$$E_{2,2} = \frac{27 \times 89}{346} = 6.96$$

$$E_{1,3} = \frac{109 \times 62}{346} = 19.53$$

$$E_{2,3} = \frac{27 \times 62}{346} = 4.47$$

$$\chi^2 = \frac{(67-61.43)^2}{61.43} + \frac{(26-15.37)^2}{15.37} + \dots = 1.884$$

$$df = (r-1)(c-1) = (2-1)(3-1) = 2$$

$$\alpha = 0.05$$

$$\chi^2_{0.05,2} = 5.991$$

$\therefore 1.884 < 5.991 \rightarrow$ Fail to reject H_0

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

Salary	Workdays at Home per Week		
	Less Than One	At Least One, but Not All	All at Home
Under \$50,000	38	16	14
\$50,000-\$74,999	54	26	12
\$75,000-\$99,999	35	22	9
Above \$100,000	33	29	12
	160	93	47

- Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.
- Use Table 5 in Appendix I to approximate the p -value for this test of hypothesis. Does the p -value confirm your conclusions from part a?

H_0 : salary & workdays at home independent

H_a : 有关

$$E_{1,1} = \frac{(18 \times 160)}{300} = 9.6$$

$$E_{1,2} = \frac{(18 \times 93)}{300} = 5.58$$

$$E_{1,3} = \frac{(18 \times 47)}{300} = 2.82$$

$$\chi^2 = 7.618$$

$$df = (4-1)(3-1) = 6$$

$$\chi^2_{0.05,6} = 12.592$$

$\therefore 7.618 < 12.592 \rightarrow$ Fail to reject H_0

when $df=6$, $\chi^2=10.595$, $p=0.1$

$\chi^2=7.618$, $p\text{-value} > 0.10$

$\rightarrow p\text{-value} > 0.05$